

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	6ab4d8aca238dc999eb5c33f
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

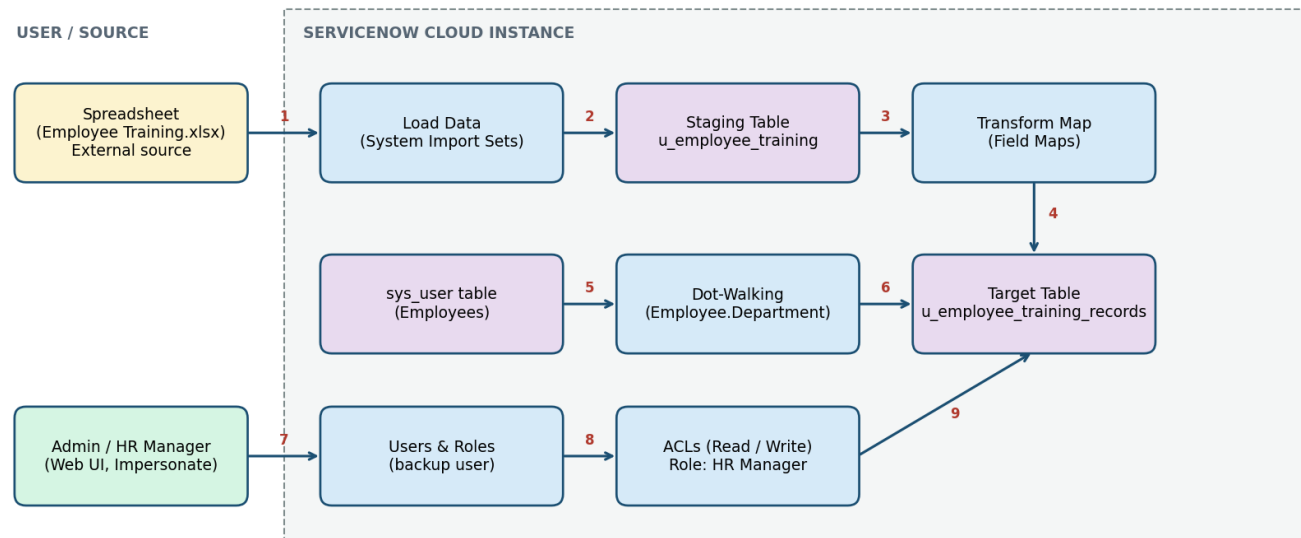


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	ServiceNow web UI used by Admin / HR Manager to create tables, load data, configure forms and view records	ServiceNow Web UI, Form Layout / List view
2.	Application Logic-1	Import of spreadsheet data into a staging table and mapping of fields to the target table	ServiceNow Import Sets (Load Data), Transform Maps
3.	Application Logic-2	Dot-walking to show the Employee's Department inside the training record	ServiceNow Form Layout Configuration (Employee.Department)
4.	Application Logic-3	Role-based access: HR Manager role, user creation, Read / Write ACLs, testing by impersonation	ServiceNow ACL, Roles, Users, Impersonate User
5.	Database	Custom table u_employee_training_records (Training Name, Completion Date, Status, Employee), staging table u_employee_training, reference to sys_user	ServiceNow relational database (Now Platform tables)
6.	Cloud Database	Database service hosted with the ServiceNow instance	ServiceNow Cloud Instance
7.	File Storage	Spreadsheet file (.xlsx) with four fields uploaded during Load Data	Local file upload (Excel / Google Sheets export)
8.	External API-1	Not applicable – data is imported from a spreadsheet file	N/A
9.	External API-2	Not applicable	N/A
10.	Machine Learning Model	Not applicable for this project	N/A

11.	Infrastructure (Server / Cloud)	Application configured and deployed on the ServiceNow instance Local Server Configuration: Not required (browser access only) Cloud Server Configuration: ServiceNow hosted instance	Cloud – ServiceNow Platform
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No external open-source framework used; built with ServiceNow low-code platform features	ServiceNow native features (Import Sets, Transform Maps, ACL)
2.	Security Implementations	Role-based access control; Read and Write ACLs restricted to the HR Manager role; security_admin elevation to create ACLs; access verified by impersonating users	ACL, Roles, Elevate Role, Impersonate User
3.	Scalable Architecture	Staging table is separate from target table, so large spreadsheets can be loaded and transformed in repeatable batches; tables and roles are reusable	ServiceNow platform, Import Sets
4.	Availability	Cloud-hosted instance accessible from any browser; availability managed by the platform	ServiceNow Cloud
5.	Performance	Records transformed in bulk by the Transform Map; dot-walked field avoids duplicate data entry	Transform Map, Dot-walking, List view